

Region-wise Causal Intervention on Epistemically Uncertain Visual Tokens for Object Hallucination in Large Vision-Language Models

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Abstract

Large vision-language models often hallucinate objects absent from the images, compromising the reliability of visual grounding. Although epistemic uncertainty correlates with hallucination, it is unclear which image regions causally drive these errors. This work presents a region-wise causal intervention framework that suppresses uncertain visual tokens in selected semantic regions, tested via a 522-image ROHE-style removed-object benchmark (COCO images with LaMa-inpainted objects) using POPE-style yes/no evaluation across five masking conditions. Applied to LLaVA-1.5-7B/13B and Qwen2.5-VL-7B, global and background masking reduced hallucination in the 7B models but not in the 13B model. Control experiments show effects aren't uniformly uncertainty-specific, and sanity checks reveal a hallucination-recognition trade-off indicating intervention effectiveness is model-dependent.

<https://github.com/PayalThangamma/region-vlm-uncertainty>

1 Introduction

Large vision-language models (LVLMs) achieve strong performance on multimodal reasoning tasks but remain prone to object hallucination, limiting their reliability in applications requiring accurate visual grounding. Recent work links epistemic uncertainty in visual representations to hallucination, showing that suppressing highly uncertain visual tokens can reduce hallucination rates. However, it remains unclear whether the observed improvements genuinely stem from uncertainty or simply from removing visual information more broadly.

RQ: *Which semantic image regions causally contribute to object hallucination in LVLMs when epistemically uncertain visual representations are selectively suppressed?*

I address this with a region-wise causal intervention framework that constructs a ROHE-style

removed-object benchmark, partitions each image into removed-object, context, and background regions, and selectively suppresses epistemically uncertain visual tokens. The resulting changes in POPE-style object-presence predictions provide a causal estimate of each region's contribution to object hallucination.

To test whether improvements are genuinely uncertainty-driven, I compare uncertainty-guided masking against matched random and matched low-uncertainty controls, and repeat all interventions on unedited images to measure the trade-off between hallucination reduction and correct recognition. I evaluate this framework on LLaVA-1.5-7B, LLaVA-1.5-13B, and Qwen2.5-VL-7B.

Global and background masking consistently reduce hallucination in LLaVA-1.5-7B and Qwen2.5-VL-7B, but not in LLaVA-1.5-13B. Uncertainty-guided masking is often statistically indistinguishable from matched random masking, suggesting scene-level visual information drives hallucination more than uncertainty localized to the removed object—i.e., the effectiveness and specificity of uncertainty-guided interventions are model-dependent, not universal. The contributions are:

- A region-wise causal intervention framework for analysing how epistemically uncertain visual representations influence hallucination across semantic image regions.
- A ROHE-style removed-object benchmark with matched random and low-uncertainty controls, original-image sanity checks, and paired bootstrap significance testing.
- Evidence that background/global evidence drives object hallucination more than removed-object or context regions, and that uncertainty-guided masking's specificity is architecture-dependent.

2 Related Work

Object hallucination has been extensively studied using benchmarks such as CHAIR (Rohrbach et al., 2018), POPE (Li et al., 2023), and ROHE (He et al., 2025). While these benchmarks provide increasingly robust protocols for evaluating hallucination, they focus on measuring hallucination rather than identifying the semantic image regions that causally contribute to it.

Seo et al. (Seo et al., 2025) linked high epistemic uncertainty in visual representations to hallucinated predictions and showed that suppressing uncertain visual tokens improves hallucination performance. However, their intervention is applied globally, leaving open whether uncertainty from different semantic regions (removed-object, context, background) contributes equally, and whether the gains stem specifically from uncertainty-guided selection or simply from removing visual information in general. No prior work has causally examined region-wise contributions under uncertainty-guided interventions, or compared them against matched random and low-uncertainty controls across multiple LLM architectures. This work addresses that gap through a region-wise causal intervention framework, motivated by the research question in Section 1.

3 Experimental Setup

3.1 Approach

I evaluate a region-wise causal intervention framework that selectively suppresses epistemically uncertain visual representations during inference. Starting from a ROHE-style removed-object benchmark, each image is partitioned into three semantic regions: the removed-object region, the surrounding context, and the remaining background. Uncertainty-guided masking is applied either globally or within one semantic region before generating object-presence predictions. The resulting change in hallucination performance estimates the causal contribution of each region. Figure 1 illustrates the construction of a ROHE-style sample and the semantic partitioning into removed-object, context, and background regions used for the proposed interventions.

3.2 Models

Experiments are conducted on three representative LLMs with different visual architectures: LLaVA-1.5-7B, LLaVA-1.5-13B, and Qwen2.5-VL-7B. For



Figure 1: Example ROHE-style sample. The original image contains the target object, while the removed-object image is generated using LaMa inpainting. The removed-object mask is converted into removed-object, context, and background token regions.

LLaVA, uncertainty masking is applied using the visual intervention proposed by Seo et al. (Seo et al., 2025). For Qwen2.5-VL, the intervention is adapted to its dynamic visual encoder while preserving the same uncertainty estimation and masking procedure.

3.3 Dataset

Experiments are conducted on a ROHE-style removed-object benchmark based on the MS COCO 2017 validation set (Lin et al., 2014; He et al., 2025). Following the ROHE methodology, target objects are removed using LaMa inpainting while preserving the surrounding scene. Images are then annotated with three semantic regions—removed-object, context, and background—to support region-wise interventions. After applying quality filtering to ensure sufficient region coverage, the final benchmark contains 522 image-question pairs spanning eight object categories. Each edited image is paired with a POPE-style binary object-presence question asking whether the removed object is still present.

3.4 Baselines

The following masking strategies are evaluated:

- **None:** no masking.
- **Global:** uncertainty-guided masking over the entire image.
- **Removed, Context, Background:** uncertainty-guided masking restricted to the corresponding semantic region.
- **Random:** matched random masking with an identical masking budget.
- **Low-Uncertainty:** matched masking of the lowest-uncertainty tokens.

To assess the impact on normal visual recognition, all masking strategies are additionally evaluated on the original, unedited images.

3.5 Evaluation

Hallucination is measured using POPE-style object-presence accuracy, where a correct prediction corresponds to identifying that the removed object is absent. For each masking condition, performance is reported as the percentage-point improvement relative to the unmasked baseline. Statistical significance is assessed using paired bootstrap resampling over image-question pairs.

3.6 Implementation Details

Following (Seo et al., 2025), epistemic uncertainty is estimated by comparing clean and adversarially perturbed visual representations. For each image, visual tokens are ranked by uncertainty, and the top- k uncertain tokens within the selected region are suppressed before decoding. The number of masked tokens is kept constant across uncertainty-guided, random, and low-uncertainty conditions to ensure fair comparison. All experiments use the official model checkpoints with identical prompting and decoding settings across masking strategies.

4 Results

This section directly answers the research question introduced in Section 1: *Which semantic image regions causally contribute to object hallucination in LVLMs under uncertainty-guided interventions?*

4.1 Which semantic regions contribute most to hallucination?

Table 1 summarizes the hallucination reduction achieved under each masking condition across the three evaluated LVLMs. Across both LLaVA-1.5-7B and Qwen2.5-VL-7B, global masking consistently produces the largest reduction in hallucination, followed closely by background masking. In contrast, masking the removed-object or context regions yields only marginal improvements or no measurable effect. LLaVA-1.5-13B does not exhibit statistically significant improvements under any masking condition.

These results directly answer the research question. Hallucination is influenced more strongly by scene-level visual information than by uncertainty localized to the removed object itself. However, this behaviour is not universal across architectures, indicating that the effectiveness of uncertainty-guided interventions remains model-dependent. Figure 2 visualizes these trends across all evaluated models and masking conditions.

Model	Global	Removed	Context	Background
LLaVA-1.5-7B	+3.64	+1.15	+0.57	+3.07
LLaVA-1.5-13B	-0.57	-0.57	-0.19	-0.57
Qwen2.5-VL-7B	+3.83	0.00	-0.19	+3.07

Table 1: Hallucination reduction (percentage points) relative to the no-masking baseline.

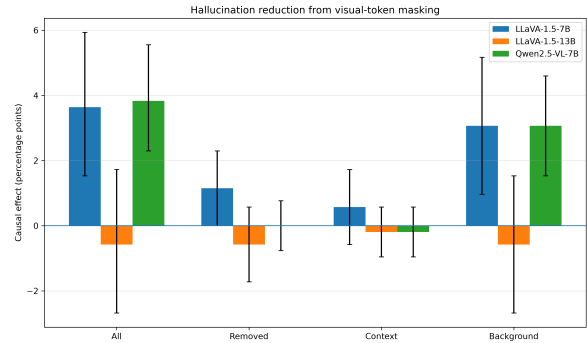


Figure 2: Hallucination reduction by model and masking condition. Positive values indicate reductions relative to each model’s no-masking baseline.

4.2 Are the improvements uncertainty-specific?

To determine whether the observed improvements are genuinely driven by epistemic uncertainty, I compare uncertainty-guided masking against matched random masking and matched low-uncertainty masking.

For LLaVA-1.5-7B, high-uncertainty masking consistently outperforms matched low-uncertainty masking, indicating that uncertainty ranking identifies visually influential representations. However, its performance is largely comparable to matched random masking, suggesting that part of the improvement can be explained by general information removal rather than uncertainty alone.

For Qwen2.5-VL-7B, high-uncertainty masking does not outperform either matched random masking or matched low-uncertainty masking. These results indicate that uncertainty specificity is backend-dependent and that uncertainty-guided masking is not universally superior to alternative masking strategies. Comparisons with matched random and matched low-uncertainty masking are provided in Appendix A.1. These controls indicate that the effectiveness of uncertainty-guided masking is architecture-dependent.

4.3 Secondary Analyses

An active-suppression analysis confirms that the weak effects observed for removed-object and con-

text masking are not caused by inactive masks. Restricting evaluation to samples with at least one suppressed token produces the same qualitative conclusions across all evaluated models.

Original-image sanity evaluation was performed to assess the trade-off between hallucination reduction and correct object recognition. Complete quantitative results are reported in Appendix A.3.

4.4 Qualitative Example

Figure 3 shows a representative case where the model hallucinates the removed bus under no-masking, removed-region, and context-region masking, all answering "Yes" despite its removal from the inpainted image. Only background-region masking corrects the answer to "No." This aligns with our aggregate results: suppressing uncertain background representations reduces hallucination more effectively than suppressing uncertain tokens localized to the object or its immediate context.

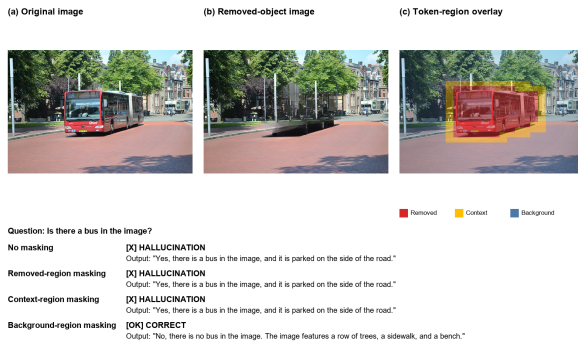


Figure 3: Qualitative example: (left) original COCO image, (center) LaMa-inpainted image with the bus removed, (right) token-level semantic region assignment. The model incorrectly predicts the bus is still present under no masking, removed-region masking, and context-region masking. Only background-region masking corrects the prediction, suggesting uncertain background information drives the hallucination in this case.

5 Conclusion

This work investigated the research question of which semantic image regions causally contribute to object hallucination in large vision-language models under uncertainty-guided interventions. To answer this question, I proposed a region-wise causal intervention framework that combines a ROHE-style removed-object benchmark with semantic region annotations and uncertainty-guided visual token suppression.

The experiments show that global and background masking consistently produce the largest

reductions in hallucination for LLaVA-1.5-7B and Qwen2.5-VL-7B, whereas masking the removed-object and context regions provides little or no benefit. In contrast, LLaVA-1.5-13B does not exhibit statistically significant improvements under any masking condition, indicating that the effectiveness of uncertainty-guided interventions depends on the underlying model architecture rather than model scale alone.

Control experiments further demonstrate that uncertainty-guided masking is not universally uncertainty-specific. While high-uncertainty masking outperforms matched low-uncertainty masking in LLaVA-1.5-7B, it does not consistently outperform matched random masking, and no clear uncertainty-specific advantage is observed in Qwen2.5-VL-7B. Together, these findings suggest that scene-level visual information contributes more strongly to object hallucination than uncertainty localized to the removed object itself.

Overall, this work extends previous uncertainty-based hallucination studies by introducing a region-wise causal perspective and stronger control experiments. The proposed framework provides a reproducible methodology for analysing where visual evidence influences hallucination and offers a foundation for future causal studies of multimodal reasoning.

Future work will extend the framework to larger and more diverse LVLMs, alternative uncertainty estimation methods, and open-ended multimodal generation tasks beyond binary object-presence evaluation.

References

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A Additional Experimental Results

This appendix presents supplementary analyses omitted from the main paper due to space limitations. These experiments provide additional evidence supporting the robustness of the proposed region-wise uncertainty-guided intervention framework.

A.1 Control Experiments

A.1.1 Matched Random Control

Matched random masking suppresses the same number of visual tokens as uncertainty-guided masking while selecting them uniformly at random within the same semantic region. Together with the matched low-uncertainty control, it evaluates whether the observed improvements are specific to epistemic uncertainty rather than general information removal. Figure 4 compares uncertainty-guided masking with the matched random and matched low-uncertainty controls.

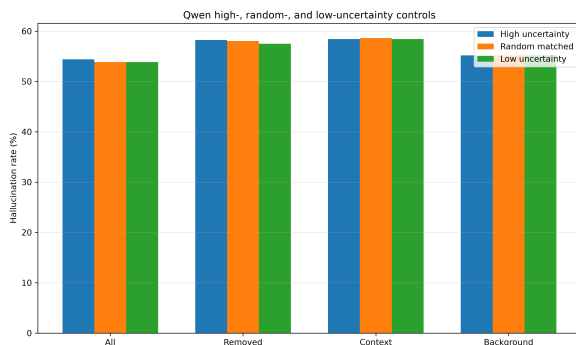


Figure 4: Comparison between high-uncertainty masking, matched random masking, and matched low-uncertainty masking for Qwen2.5-VL-7B.

A.1.2 Matched Low-Uncertainty Control

As an additional control, the same masking budget is applied to the lowest-uncertainty visual tokens. For LLaVA-1.5-7B, high-uncertainty masking consistently outperforms low-uncertainty masking, whereas Qwen2.5-VL-7B exhibits little difference between the two strategies. These results indicate that uncertainty-guided masking is not uniformly uncertainty-specific across architectures.

A.2 Original-Image Sanity Evaluation

The masking strategies are additionally evaluated on the original, unedited images where the queried object is present. These experiments quantify the trade-off between hallucination reduction and preservation of correct visual recognition.

A.3 Active Suppression Ablation

To verify that the weak removed-object and context-region effects are not caused by inactive masks, the analysis is repeated using only samples containing at least one suppressed visual token. The qualitative conclusions remain unchanged across all evaluated models, indicating that the limited improvements for these regions cannot be explained by empty intervention regions.

A.4 Statistical Significance

Tables 3–5 report the complete paired bootstrap confidence intervals and corresponding bootstrap p-values for all evaluated models.

Condition	Effect (pp)	95% CI	<i>p</i>
All	+3.64	[1.53, 5.94]	0.0014
Removed	+1.15	[0.00, 2.30]	0.0642
Context	+0.57	[-0.57, 1.72]	0.3840
Background	+3.07	[0.96, 5.17]	0.0036

Table 3: Paired bootstrap results for LLaVA-1.5-7B.

Condition	Effect (pp)	95% CI	<i>p</i>
All	-0.57	[-2.87, 1.53]	0.6588
Removed	-0.57	[-1.72, 0.57]	0.4016
Context	-0.19	[-1.15, 0.57]	0.8472
Background	-0.57	[-2.68, 1.53]	0.6800

Table 4: Paired bootstrap results for LLaVA-1.5-13B.

Model	Condition	Accuracy (%)	Drop (pp)	95% CI	<i>p</i>
LLaVA-1.5-7B	None	94.64	0.00	–	–
LLaVA-1.5-7B	All	92.34	2.30	[0.77, 4.02]	0.0052
LLaVA-1.5-7B	Removed	94.25	0.38	[-0.38, 1.15]	0.4550
LLaVA-1.5-7B	Context	94.25	0.38	[0.00, 0.96]	0.2760
LLaVA-1.5-7B	Background	92.72	1.92	[0.38, 3.45]	0.0136
Qwen2.5-VL-7B	None	90.42	0.00	–	–
Qwen2.5-VL-7B	All	87.74	2.68	[1.15, 4.41]	0.0002
Qwen2.5-VL-7B	Removed	90.23	0.19	[-0.38, 0.77]	0.7892
Qwen2.5-VL-7B	Context	90.42	0.00	[-0.57, 0.57]	1.0000
Qwen2.5-VL-7B	Background	88.51	1.92	[0.77, 3.26]	0.0016

Table 2: Original-image sanity evaluation. Positive drops indicate reduced correct recognition after masking.

Condition	Effect (pp)	95% CI	<i>p</i>
All	+3.83	[2.30, 5.56]	< 0.0001
Removed	0.00	[-0.77, 0.77]	1.0000
Context	-0.19	[-0.96, 0.57]	0.8352
Background	+3.07	[1.53, 4.60]	< 0.0001

Table 5: Paired bootstrap results for Qwen2.5-VL-7B.

Condition	Yes	No	Unk.	Halluc. (%)	Effect
None	304	218	0	58.24	0.00
All	284	238	0	54.41	+3.83
Removed	304	218	0	58.24	0.00
Context	305	216	1	58.43	-0.19
Background	288	234	0	55.17	+3.07

Table 8: Qwen2.5-VL-7B hallucination results. Effect values are percentage points.

A.5 Detailed Per-Model Results

The complete quantitative results for LLaVA-1.5-7B, LLaVA-1.5-13B, and Qwen2.5-VL are provided in this section for completeness. These detailed tables complement the summarized comparisons presented in the main paper.

Condition	Yes	No	Halluc. (%)	Effect (pp)
None	415	107	79.50	0.00
All	396	126	75.86	+3.64
Removed	409	113	78.35	+1.15
Context	412	110	78.93	+0.57
Background	399	123	76.44	+3.07

Table 6: LLaVA-1.5-7B hallucination results on 522 removed-object samples.

Condition	Yes	No	Halluc. (%)	Effect (pp)
None	463	59	88.70	0.00
All	466	56	89.27	-0.57
Removed	466	56	89.27	-0.57
Context	464	58	88.89	-0.19
Background	466	56	89.27	-0.57

Table 7: LLaVA-1.5-13B hallucination results.

B Disclaimer: Use of AI

ChatGPT was used to assist with language editing, improving readability, and refining the structure of the manuscript. All research ideas, experimental design, implementation, analysis, interpretation of results, and final scientific conclusions were developed and verified by the author.